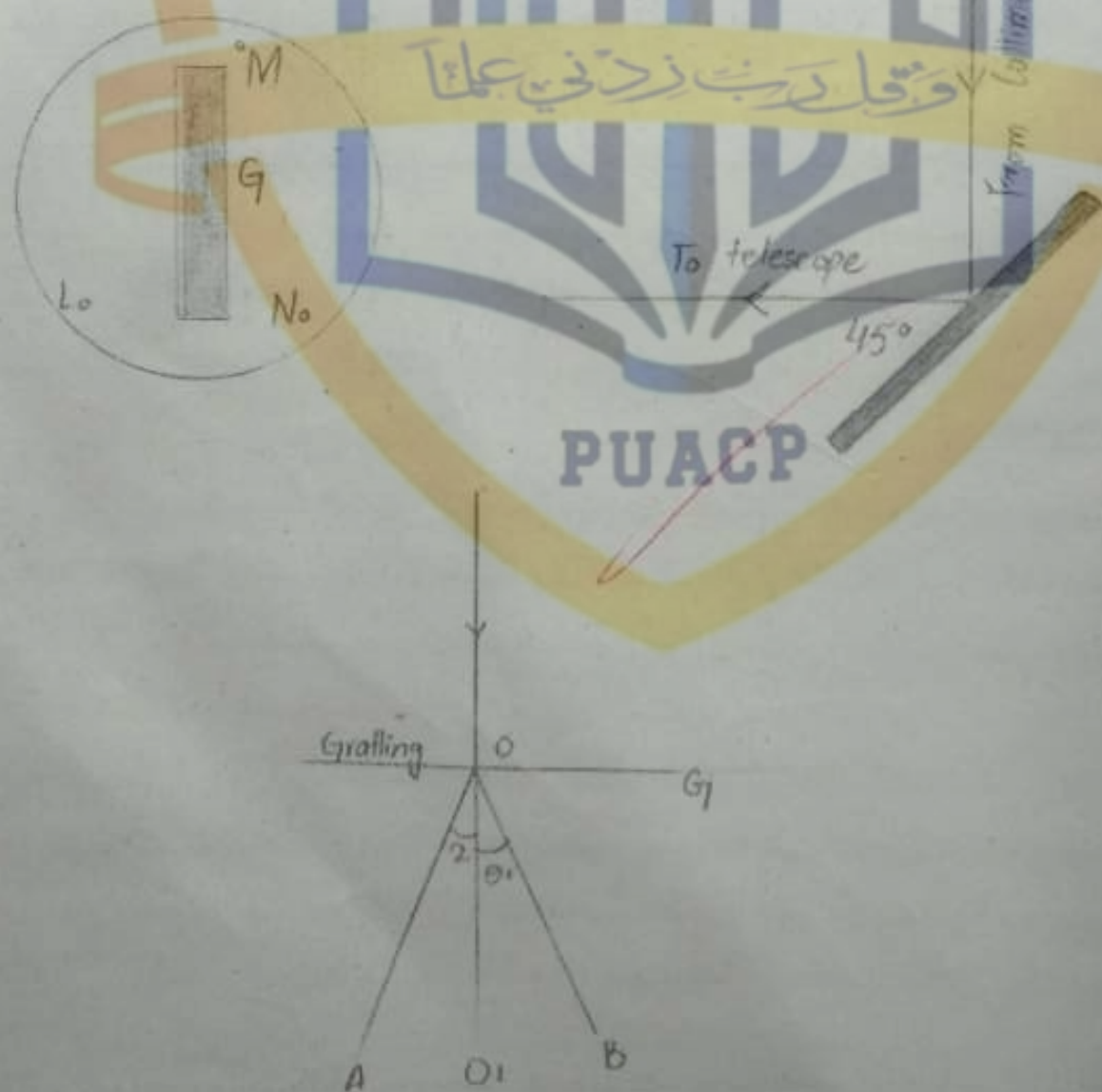


Experiment :- To determine the wavelength of Sodium light by a plan diffraction grating, using Spectrometer.

Apparatus:

- A spectrometer
- Spirit level
- Sodium lamp
- diffraction grating with clamping arrangement eye-piece

Diagram:



Experiment Name: To determine the wavelength of sodium light by a plan diffraction grating, using spectrometer

Apparatus :-

- A spectrometer
- Spirit level
- Sodium lamp
- diffraction grating with clamping arrangement
- eye-piece.

Procedure :-

- Focus the eye-piece on cross-wire and focus the telescope and collimator for parallel rays of light. level the prism table with the help of spirit level.
- Turn the telescope to left through 90° .
- Fix the telescope. Place the grating on prism table and make sure its perpendicular to the line joining two of the levelling screws.
- Read the vernier and rotate the mirror through 45° in the direction of arrow head.
- Adjust the telescope to see the clear image and note vernier position of the direct image.
- Now turn the telescope to the right of the central position.
- Read the number's of lines per cm.

Observation and Calculation:

$$\text{Least Count of Spectrometer} = \frac{\text{Value of smallest scale division}}{\text{No. of division on Vernier Scale}}$$

$$= \frac{1^\circ}{10} = 6 \text{ min}$$

$$\text{No. of line rule on grating} = 6000 / \text{cm}$$

$$\text{Grating Element } (a+b) = \frac{1}{6000} = 1.67 \times 10^{-4} \text{ cm}$$

Order of Spectrum	Vernier	Reading For			Angle of diffraction			$\sin \theta$	Wavelength $\lambda = \frac{(a+b)\sin \theta}{n}$
		Left	Direct	Right	Left	Right	Mean		
1 st Order $n=1$	V_1	$117^\circ 30'$	$96^\circ 42'$	$75^\circ 56'$	$20^\circ 48'$	$20^\circ 36'$	$20^\circ 42'$	$\sin(20^\circ 42')$ 0.3194	$\frac{(1.67 \times 10^{-4})(0.3194)}{1}$ 5.334×10^{-5}
2 nd Order $n=2$	V_2	$142^\circ 54'$	$96^\circ 42'$	$52^\circ 66'$	$46^\circ 12'$	$43^\circ 56'$	$44^\circ 54'$	$\sin(44^\circ 54')$ 0.6483	$\frac{(1.67 \times 10^{-4})(0.6483)}{2}$ 5.413×10^{-5}

$$\text{Mean Wavelength} = \frac{5.334 \times 10^{-5} + 5.413 \times 10^{-5}}{2} = 5.373 \times 10^{-5} \text{ cm}$$

$$= 5.373 \times 10^{-5} \times 6.684 \times 10^{-12} = 3.59 \times 10^{-16} \text{ AU}$$

$$\text{Result: Wavelength of Sodium light} = 5373.0 \text{ \AA}, 5.373 \times 10^{-5} \text{ cm}$$

$$\text{Actual Value of } \lambda = 5896 \times 10^{-8} \text{ cm} = 5.896 \times 10^{-5} \text{ cm}$$

$$\text{Percentage error} = \frac{\text{Actual Value} - \text{Calculated Value}}{\text{Actual value}} \times 100$$

$$= \frac{5.896 \times 10^{-5} - 5.373 \times 10^{-5}}{5.896 \times 10^{-5}} \times 100 = 0.0887 \times 100 = 8.87 \%$$

let the number be 6000 cm then
grating element $a+b = \frac{1}{6000}$ cm

- In formula $(a+b)\sin\theta = n\lambda$, the value of n will be 1 and 2 for 1st and 2nd order spectrum respectively.

بقلم: محمد علي

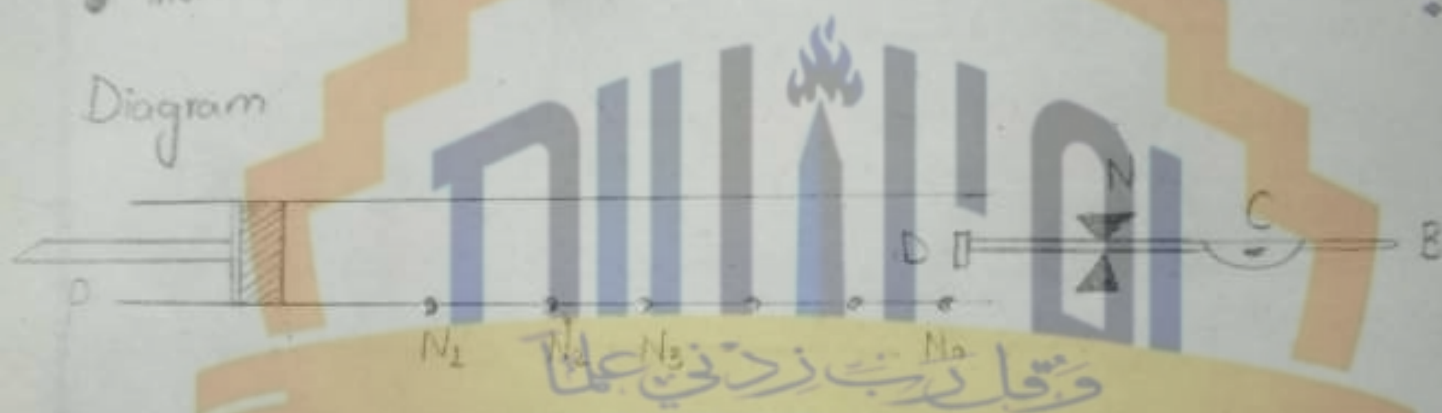
PUACP

Object: To determine the velocity of sound by Kundt's tube.

Apparatus:

- Kundt's tube fitted with a metal rod
- lycopodium powder
- a piece of resined leather
- Thermometer
- meter rod

Diagram



Observation and Calculation:-

Room temperature = $t = 27^\circ \text{C}$

Density of material of rod = $\rho = 4.8 \text{ gm-cm}^{-3}$

Young Modulus for Rod = $\gamma = 9.7 \times 10^{11} \text{ dynes-cm}^{-2}$

Length of Rod = $DB = L = 57 \text{ cm}$

To determine the velocity of sound by Kundt's tube

Apparatus :

- Kundt's tube fitted with a metal rod
- Lycopodium powder
- Thermometer
- a piece of resined leather
- Meter rod

Procedure :-

- Take a Kundt's tube which is a glass tube about 100 cm long 3.5 cm in diameter.
- Place the tube horizontally on V-shaped supports on the table and sprinkle a fine layer of lycopodium powder inside the tube.
- Close one end with the movable piston P inside the tube at one end and metal rod carrying the chark disc D at the other end.
- Set up longitudinal vibration in the rod by rubbing it with resined leather. At the same time position of piston in the tube such that the lycopodium powder is violently agitated when air column resonates with the note of rod.
- Measure the distance L from the

No. of Obs.	Total distance b/w the first node and last 'l' cm	No. of spaces in total distance L 'n'	Average distance b/w two nodes $\lambda_0 = \frac{l}{n}$ cm	Velocity of sound in air $V_n = \sqrt{\frac{\gamma}{\rho}} \cdot \frac{\lambda_0}{L_n}$
1.	25	4	$\lambda_0 = \frac{25}{4} = 6.25$	$V_n = \sqrt{\frac{9.7 \times 10^4}{4.8}} \cdot \frac{6.25}{57} = 49291.3$
2.	10	4	$\lambda_0 = \frac{10}{4} = 2.5$	$V_n = \sqrt{\frac{9.7 \times 10^4}{4.8}} \cdot \frac{2.5}{57} = 19716.5$

Mean Velocity of sound in air at room temperature = $\frac{49291.3 + 19716.5}{2} = 34503.9 \approx 34504 \text{ cm-sec}^{-1}$

Result: (i) Velocity of sound in air at room temperature = 34504 m-sec^{-1}

(ii) Velocity of sound at $0^\circ\text{C} = V_0 = \sqrt{\frac{T_0}{T}}$

$$V_0 = V_n \sqrt{\frac{273}{273+t}} = 34504 \sqrt{\frac{273}{273+27}} = 34504 \times 0.953$$

$$V_0 = 32914.7 \text{ cm-sec}^{-1}$$

Correct Value of $V_0 = 347.3 \text{ m-sec}^{-1}$ at 27°C
 $= 34730 \text{ cm-sec}^{-1}$

Error = $\frac{\text{Correct Value} - \text{Calculated Value}}{\text{Correct Value}} \times 100$

$$= \frac{34730 - 32914.7}{34730} \times 100 = 5.2 \%$$

middle of 1st heap N_1 to the middle of N^{th} heap N_n .

- Measure the length L_n of rod and note down the temperature.
- Find out the velocity of sound in air at room temperature by using

$$V_t = \sqrt{\frac{Y}{R} \cdot \frac{L_n}{L_n}}$$

- Repeat the experiment twice and find out the reading.
- The velocity of sound at 0°C can be calculated by

$$V_0 = V_t \sqrt{\frac{273}{273 + 27}}$$